

Artin 4

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February 28, 2024

Problem 6.4.

Proof. By last semester, $O \cong S_4$. The conjugacy classes are denoted by the partition of n elements. So $[1]$ is trivial, $[2 + 1 + 1]$ is a two cycle, $[2 + 2]$ is a double 2 cycle, $[3 + 1]$ is a three cycle, and $[4]$ is a four cycle.

By last semester, $|[1]| = 1$, $|[2 + 1]| = \binom{4}{2} = 6$, $|[2 + 2]| = \frac{4!}{2^2 2!} = 3$, $|[3 + 1]| = \frac{4!}{[3*1]*1} = 8$, and $[4] = 6$.

The corresponding rotations for each conjugacy class are the following:

1. $[1] : I$
2. $[2 + 1 + 1] : R_{F,\pi}$, rotation through face by π radians, because these rotations are 2 cycles.
3. $[2 + 2] : R_E$, edge rotations
4. $[3 + 1] : R_V$, vertex rotations fix 1 diagonal and cycle the other 3 diagonals.
5. $[4] : R_{F,\frac{\pi}{2}}$, rotation through face by $\frac{\pi}{2}$ radians have order 4 and cycle all diagonals.

χ_{faces} is the character of the permutation representation which permutes all faces. χ_{FP} , is the character for the permutation representation which permutes all face pairs (opposite faces). χ_V , is the character for the permutation representation which permutes set of vertices. χ_D , standard permutation representation of S_n i.e permutation representation which permutes diagonals. χ_E , permutation representation which permutes opposite edges. χ_{tets} , permutation representation which permutes the inscribed tetrahedron.

	(1)	(3)	(6)	(6)	(8)
	1	$R_{F,\pi}$	R_E	$R_{F,\frac{\pi}{2}}$	R_V
	1	$2 + 2$	$2 + 1 + 1$	4	$3 + 1$
χ_{faces}	6	2	0	2	0
χ_{FP}	3	3	1	1	0
χ_V	8	0	0	0	2
χ_D	4	0	2	0	1
χ_E	6	0	2	0	0
χ_{tets}	2	2	0	0	2

	(1)	(3)	(6)	(6)	(8)
	1	R_E	$R_{F,\pi}$	$R_{F,\frac{\pi}{2}}$	R_V
	1	2 + 2	2 + 1 + 1	4	3 + 1
χ_{triv}	1	1	1	1	1
χ_{sign}	1	1	-1	-1	1
$\chi_3 = \chi_{FP} - \chi_{triv}$	2	2	0	0	-1
$\chi_4 = \chi_D - \chi_{triv}$	3	-1	1	-1	0
$\chi_5 = \chi_4 * \chi_{sign}$	3	-1	-1	1	0

All the self inner products are 1 and the number of characters are equal to the number of conjugacy classes. $\square$

Problem 6.5.

Proof. $\forall n \in \mathbb{N}$, $\mathbb{C}_{triv} = \text{span}(e_1 + \dots + e_n)$ because any permutation matrix fixes $\begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}$ and any

scalar multiples of it. So start by decomposing $\mathbb{C}^n = \mathbb{C}_{triv} \oplus C_{triv}^\perp$, where $C_{triv}^\perp$ are the set of vectors perpendicular under the hermitian dot product to $e_1 + \dots + e_n$. Thus $\forall v \in C_{triv}^\perp$, $\sum_{i=1}^n v_i = 0$. Let $W \leq C_{triv}^\perp$, suppose W is S_n -invariant. $\forall v \in W \ni v \neq 0$, any linear combination of v and the permutation of the components of v is in W because S_n are permutation matrices. Moreover all the components of v are not equal because then $v \in \mathbb{C}_{triv}$, but $C_{triv} \cap C_{triv}^\perp = 0$. Since $v \in W$ and its permutations are in W , permute v such that the order of its components are descending and redefine $v = (v_1, \dots, v_n)$ to such a permutation (where $v_i \geq v_j$ if $i \leq j$) then $v' = (v_2, v_1, v_3, \dots, v_n) \in W$, since permutations of $v \in W$. $v - v' = (v_1 - v_2, v_2 - v_1, 0, \dots, 0) \in W$, then $\frac{v-v'}{v_1-v_2} = (1, -1, 0, \dots, 0) = e_1 - e_2 \in W$. With a similar argument (taking a different permutation) you can get all $e_1 - e_j \in W$ for all $j > 1$. $\forall a_2, \dots, a_n \in \mathbb{C}$, $0 = \sum_{j=2}^n a_j(e_1 - e_j) = e_1(n-1) \sum_{j=2}^n a_j + \sum_{j=2}^n (-a_j e_j)$, $\{e_i\}_i$ is the standard basis and thus is linear independent thus $a_j = 0$ and $\sum_{j=2}^n a_j = 0$. Thus $\{e_1 - e_j\}_{1 < j \leq n}$ is linearly independent. $\dim \text{span}\{e_1 - e_j\}_{1 < j \leq n} = n - 1 \leq \dim \mathbb{C}_{triv}^\perp = n - 1$ and $\text{span}\{e_1 - e_j\}_{1 < j \leq n} \leq W \leq C_{triv}^\perp$, thus $\mathbb{C}_{triv}^\perp = \text{span}\{e_1 - e_j\}_{1 < j \leq n} = W$. Thus $C_{triv}^\perp$ is the only S_n invariant proper subspace of $C_{triv}^\perp$, thus $C_{triv}^\perp$ is irreducible. $\square$

Problem 6.6.

Proof. $I \cong A_5$. Here are the elements wrt to I, R_F (rotation through face), R_E (rotation through edge), R_{V1} (rotation of 72 degrees through vertices), R_{V2} (rotation of 144 degrees through vertices).

By last semester, $|[I]| = 1$, $|[R_F]| = 20$, $|[R_E]| = 15$, $|[R_{V1}]| = |[R_{V2}]| = 12$.

	(1)	(12)	(12)	(15)	(20)
	I	R_{V1}	R_{V2}	R_E	R_F
χ_F	20	0	0	0	2
χ_V	12	2	2	0	0
χ_E	30	0	0	2	0

From Artin's section on the character table of $I \cong A_5$. Let $\alpha = 2 \cos(2\pi/5) + 1$ and $\beta = 2 \cos(4\pi/5) +$

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	(1)	(12)	(12)	(15)	(20)
	1	[5]	[5]	[2 + 2 + 1]	[3 + 1]
	I	R_{V1}	R_{V2}	R_E	R_F
χ_1	1	1	1	1	1
χ_2	3	α	β	-1	0
χ_3	3	β	α	-1	0
χ_4	4	-1	-1	0	1
χ_5	5	0	0	1	-1

Use the inner product to determine the multiplicities of the irreducibles.

1. $\langle \chi_F, \chi_1 \rangle = \frac{1}{60} (20 + 40) = 1$. $\langle \chi_F, \chi_2 \rangle = \frac{1}{60} (60) = 1$. $\langle \chi_F, \chi_3 \rangle = 1$. $\langle \chi_F, \chi_4 \rangle = \frac{1}{60} (80 + 40) = 2$.
 $\langle \chi_F, \chi_5 \rangle = \frac{100-40}{60} = 1$. Thus $\chi_F = \chi_1 + \chi_2 + \chi_3 + 2\chi_4 + \chi_5$. Thus $V_F = V_1 \oplus V_2 \oplus V_3 \oplus V_4^2 \oplus V_5$
2. $\langle \chi_V, \chi_1 \rangle = \frac{1}{60} (12 + 24 + 24) = 1$. $\langle \chi_V, \chi_2 \rangle = \frac{1}{60} (36 + 24\alpha + 24\beta) = \frac{1}{60} (36 + 24 + 24 - 24) = 1$.
 $\langle \chi_V, \chi_3 \rangle = 1$. $\langle \chi_V, \chi_4 \rangle = \frac{1}{60} (48 - 24 - 24) = 0$. $\langle \chi_V, \chi_5 \rangle = \frac{60}{60} = 1$. Thus $\chi_V = \chi_1 + \chi_2 + \chi_3 + \chi_5$.
Thus $V_V = V_1 \oplus V_2 \oplus V_3 \oplus V_5$
3. $\langle \chi_E, \chi_1 \rangle = \frac{1}{60} (30 + 30) = 1$. $\langle \chi_E, \chi_2 \rangle = \frac{1}{60} (90 - 30) = 1$. $\langle \chi_E, \chi_3 \rangle = 1$. $\langle \chi_E, \chi_4 \rangle = \frac{1}{60} (120) = 2$.
 $\langle \chi_E, \chi_5 \rangle = \frac{150+30}{60} = 3$. Thus $\chi_E = \chi_1 + \chi_2 + \chi_3 + 2\chi_4 + 3\chi_5$. Thus $V_E = V_1 \oplus V_2 \oplus V_3 \oplus V_4^2 \oplus V_5^3$

□

Problem 6.7.

Proof. From 4.12, we know that $\forall \sigma \in A_n$, conjugation by σ doesn't change the conjugacy classes since $\sigma \in A_n$ and thus characters don't change. $\forall \sigma \notin S_n$ we know that some conjugacy classes may be permuted depending on their sizes, from last semester. From last semester we know the class equation of A_5 in S_5 is $1 + 24 + 15 + 20$ where 24 corresponds to the conjugacy class of the 5 cycles, 15 is the double transpositions cycles, and 20 is the 3 cycles. Thus when we restrict to A_5 we know that the conjugacy class of 5 cycles is the only one to break apart, and conjugation of a 5 cycle by α produces a 5 cycle in the other conjugacy class. Thus conjugation by sigma swaps the characters of the 2 conjugacy class of the 5 cycles. This only effects the dimension 3 characters, as all other characters have the same value for both conjugacy classes. Thus the resultant character table is:

	(1)	(12)	(12)	(15)	(20)
	1	[5]	[5]	[2 + 2 + 1]	[3 + 1]
$\chi_1 \circ C_\sigma$	1	1	1	1	1
$\chi_2 \circ C_\sigma$	3	β	α	-1	0
$\chi_3 \circ C_\sigma$	3	α	β	-1	0
$\chi_4 \circ C_\sigma$	4	-1	-1	0	1
$\chi_5 \circ C_\sigma$	5	0	0	1	-1

This swap of the resultant values preserves all the characters corresponding to the irreducibles of dimension other than three. However this swaps the irreducibles with dimension three as $\chi_2 \circ C_\sigma = \chi_3$ and $\chi_3 \circ C_\sigma = \chi_2$. Thus the conjugation action swaps the iso classes of irreducibles with characters corresponding to χ_2 and χ_3 and fixes the other iso classes. Thus the conjugation action of S_5 on the isomorphic classes of irreducibles of A_5 either swap the 3 dimensional irreducibles or fix everything. □

Problem 6.10.

Proof. The only non abelian groups of order 12, are A_4 , D_6 , and $\mu_3 \rtimes \mu_4$.

The conjugacy classes of A_4 are $[1], [2+2], [3+1]_a, [3+1]_b$, where a and b denote the different 3 cycle conjugacy classes. The sizes are 1, 3, 4, 4 respectively. K_4 is the only normal subgroup of A_4 and $A_4/K_4 \cong \mu_3$, thus $K_4 \cong [A_4, A_4]$. There are 3 maps from $\mu_3 \rightarrow \mu_3 \leq C^*$, trivial map, identity map, and inversion map. Each map corresponds to one of the 1 dimensional representations. Thus the dimension of the last representation is $\sqrt{12-3(1)} = \sqrt{9} = 3$. χ_{stan} is the standard representation. $\chi_4 = \chi_{stan} - \chi_1$ and the self inner of $\langle \chi_4, \chi_4 \rangle = \frac{1}{12}(3*3+3(1)) = 1$, thus χ_4 is irreducible. Let $\omega = e^{\frac{2\pi}{3}}$

	(1) 1	(3) [2+2]	(4) [3+1] _a	(4) [3+1] _b
χ_1	1	1	1	1
χ_2	1	1	ω	$\bar{\omega}$
χ_3	1	1	ω	$\bar{\omega}$
χ_4	3	-1	0	0
χ_{reg}	12	0	0	0
χ_{stan}	4	0	1	1

The character table for D_6 was calculated in problem 4.3 in Artin 2.

From the irreducible representations of D_n homework, we know the 2 dimension irreducibles of D_6 have the following form. $x \rightarrow \begin{bmatrix} \omega^k & \\ & \omega^{-k} \end{bmatrix}$ where $\omega = e^{\pi i/3}$ (6th root of unity) and $k \in \{1, 2\}$ and

$y \rightarrow \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. In either representation, the trace of y and xy is 0. $tr(x) = \omega^k + \omega^{-k} = 2\cos(k\frac{\pi}{3}) = (-1)^{k+1}\sqrt{3}$

	e	x^2	x	y	xy
χ_1	1	1	1	1	1
χ_2	1	1	-1	1	-1
χ_3	1	1	1	-1	-1
χ_4	1	1	-1	-1	1
χ_5	2	-2	$\sqrt{3}$	0	0
χ_6	2	-2	$-\sqrt{3}$	0	0
χ_{reg}	12	0	0	0	0

The character table for $G = \mu_3 \rtimes \mu_4 = \langle x, y | x^3 = y^4 = 1, yxy^{-1} = x^{-1} \rangle$ was determined for 4.9 from Artin 3. Here is my derivation of it, $\mu_3 \triangleright G$ and $xyx^{-1}y^{-1} = x^{-2} = x$. Thus $\langle x \rangle = \mu_3 \leq [G, G]$. But $G/\mu_3 = \mu_4$ which was abelian, thus $\mu_3 = [G, G]$. Thus all endomorphisms for μ_4 give all the 1 dimensional representations of G .

Let V be a 2 dimensional representation. Since $x^3 = 1$, x is diagonalizable with eigenvalues being 3rd roots of unity. Let λ be a eigenvalue and v be a corresponding eigenvector.

$xy^{-1}v = yy^{-1}xyv = y^{-1}x^{-1}v = \bar{\lambda}y^{-1}v$. Thus $\bar{\lambda}$ is a eigenvalue of x with $y^{-1}v$ being the corresponding eigenvector.

Thus $x = \begin{bmatrix} \lambda & 0 \\ 0 & \bar{\lambda} \end{bmatrix}$ and $y^{-1} = y = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. $\lambda \notin \mathbb{R}$, because then $span(v + y^{-1}v)$ is G -invariant. Thus let $\lambda = e^{2\pi i/3}$. $tr(x) = \lambda + \bar{\lambda} = 2\cos(2\frac{\pi}{3}) = -1$ and $tr(y) = 0$. This representation is irreducible, because $\langle \chi_5, \chi_5 \rangle = \frac{1}{12}(4+4+2+2) = 1$. Now to get χ_6 use the regular representation

	(1) 1	(1) y^2	(2) x	(2) xy^2	(3) y	(3) y^{-1}
χ_1	1	1	1	1	1	1
χ_2	1	1	1	1	-1	1
χ_3	1	-1	1	-1	i	-i
χ_4	1	-1	1	-1	-i	i
χ_5	2	2	-1	-1	0	0
χ_6	2	-2	-1	1	0	0
χ_{reg}	12	0	0	0	0	0

□